

OSSP-2

Question:

2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

Solution:

```
GNU nano 8.7.1 practical2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main()
{
    int source, destination;
    char buffer[1024];
    int bytesRead;

    source = open("source.txt", O_RDONLY);

    if(source < 0)
    {
        printf("Unable to open source file\n");
        return 1;
    }

    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC,
                      0644);

    if(destination < 0)
    {
        printf("Unable to create destination file\n");
        close(source);
        return 1;
    }

    while((bytesRead = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytesRead);
    }

    close(source);
    close(destination);

    printf("File copied successfully.\n");

    return 0;
}
```

Source.txt

```
GNU nano 8.7.1 source.txt
Operating Systems Lab
Practical 2
Using open read write and close
```

```
ubuntu@ubuntu:~$ nano practical2.c
ubuntu@ubuntu:~$ nano source.txt
ubuntu@ubuntu:~$ gcc practical2.c -o practical2
ubuntu@ubuntu:~$ ./practical2
File copied successfully.
```

Optional features enabled: no-m32-mpers

```
ubuntu@ubuntu:~$ strace cat sample.txt
```

```
ubuntu@ubuntu:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0xffffdc6577e8 /* 53 vars */) = 0
brk(NULL)                               = 0xbbe1dbd30000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0xe2dc32fa0000
faccessat(AT_FDCWD, "/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=72069, ...}) = 0
mmap(NULL, 72069, PROT_READ, MAP_PRIVATE, 3, 0) = 0xe2dc32f50000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/aarch64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=264608, ...}) = 0
mmap(NULL, 337280, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_DENYWRITE, -1, 0) = 0xe2dc32f05000
mmap(0xe2dc32f10000, 271664, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xe2dc32f10000
munmap(0xe2dc32f05000, 45056)            = 0
munmap(0xe2dc32f53000, 17712)           = 0
mprotect(0xe2dc32ef0000, 61440, PROT_NONE) = 0
mmap(0xe2dc32f4f000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3f000) = 0xe2dc32f4f000
mmap(0xe2dc32f51000, 5424, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0xe2dc32f51000
mmap(0xe2dc32f10000, 196608, PROT_READ|PROT_EXEC|0x10, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xe2dc32f10000
close(3)                                = 0
openat(AT_FDCWD, "/usr/lib/aarch64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0\267\0\1\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=133776, ...}) = 0
mmap(NULL, 263072, PROT_NONE, MAP_PRIVATE|MAP_ANONYMOUS|MAP_DENYWRITE, -1, 0) = 0xe2dc32ecf000
mmap(0xe2dc32ed0000, 197536, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xe2dc32ed0000
munmap(0xe2dc32ecf000, 4096)            = 0
munmap(0xe2dc32f01000, 50272)           = 0
mprotect(0xe2dc32ee0000, 65536, PROT_NONE) = 0
mmap(0xe2dc32eff000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1f000) = 0xe2dc32eff000
mmap(0xe2dc32ed0000, 126976, PROT_READ|PROT_EXEC|0x10, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0) = 0xe2dc32ed0000
```